Exploiting Basins of Attraction for Newton's Method on Brockett Function Maximization

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Abstract

This work provides novel attempts in incorporating the knowledge to basins of attraction into optimization algorithm design. This work analyzes the geometry and the symmetries to the basins of attraction generated by the Newton's method on the Brockett function on $\mathrm{SO}(m)$. Using this knowledge, we design two optimization schemes using the Newton's method that ensure convergence to the global maxima. The first scheme combines gradient ascent and Newton's method for maximizing the Brockett function, optimal step size and number of iterations for the gradient ascent is given. The second scheme provides a suitable initialization set for the method of population, explicit method on $\mathrm{SO}(2)$ is given. The results motivate future endeavors in utilizing the knowledge to the basins of attraction under iteration maps to provide faster and more robust optimization algorithms.

Keywords: 7 or fewer keywords

1 Introduction

This work focuses on the following constrained maximization problem:

$$\max_{X\in\mathrm{SO}(m)} f(X) = \max_{X\in\mathrm{SO}(m)} \mathrm{Tr}\left\{X^\mathsf{T} A X B\right\}, \tag{1}$$

where $\mathrm{SO}(m)$ is the set of $m\times m$ special orthogonal matrices, A and B are symmetric matrices each haiving no repeating eigenvalues (i.e., all eigenvalues have geometric multiplicity equal to one), and $f(X) = \mathrm{Tr}\left\{X^\mathsf{T} A X B\right\}$ is termed the Brockett function. R.W. Brockett discussed in great lengths the maximization of this function, and he utilized its maximization process in sorting lists,

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diagonalizing symmetric matrices, and solving linear programming problems [1]. More recently, the Brockett function is also related to the C-numerical range from quantum control theory, and it optimization can be used to describe distances between quantum states [7].

Standard ascent methods used to maximize a function include the gradient ascent method and the Newton's method. These methods gives an iteration map at each point, providing an ascent direction for the variable. It has long been observed that even the simplest iterative maps can result in non-trivial chaotic behaviors [8, 9]. The study of chaotic dynamical systems has long been established, focusing especially on holomorphic dynamics, which are complex analytic mappings such as using Newton's method for finding zeros to a polynomial. The domain of the map is divided into different basins of attraction, where given an initial point at one of the basins decides which critical point it converges to deterministically. The boundaries to the basins are usually fractals, where a small perturbation to the initial condition can result in drastically different output dynamics. Such dynamics are chaotic. Our work is the first to analyze geometry to the basins of attraction of the Brockett function on the manifold $\mathrm{SO}(m)$.

Furthermore, as far as we have seen, our work is the first to apply the geometry of the basins of attraction to optimization algorithm design. While using the Newton's method for optimization, the impact of chaotic dynamics greatly impedes the performances of the algorithms [cite]. However, none of the previous methods related to Newton's method utilizes the knowledge to its basins of attraction as a resource for optimization methods designing. This is obvious since the fractal structures to the basins vary drastically given different parameters to functions. The results are often case-specific and cannot be further generalized or utilized to other functions. The Brockett function, however, has inherent coordinate-free symmetries to its formulation. And by considering also a coordinate-free Newton's method, the symmetry translates to the geometry of the basins of attraction, no matter what the parameters to the Brockett function is. The above properties make the Brockett function a good choice for a further study.

Using the knowledge to the Brockett function, we propose two novel optimization schemes for the maximization to the Brockett function. The first scheme iterates gradient ascent followed by Newton's method. This is a usual scheme to reduce the chaotic behavior in Newton's method by first traversing into a basin of attraction and away from the fractal boundaries [cite]. However, we are the first to provide methods to analyze the optimal step size α and number of iterations

n needed for the gradient ascent to ensure that, given a random initial condition, one has the highest probability in converging to the global maxima. The second scheme is a realization to the method of population, where multiple initial conditions are given to avoid converging to local extrema. Given the structure to the basins of attraction, we can in fact design a deterministic arrangement of initial condition that ensure convergence to global maxima with probability one.

Our main results include: the analysis to the geometry of basins of attraction of the Brockett function under the Newton's method (Section 3), optimal step size and iterations needed for gradient ascent followed by Newton's method (Section 4), and arrangement of initial points for population methods (Section 5). All of which shows that this new approach in studying the fractal structures can be used as a resource in optimization algorithm designing.

2 Ascent Direction Methods on the Brockett Function

Traditionally, maximization can be realized by iterating the ascent direction methods. Two renowned ascent direction methods for maximizing a function are the gradient ascent method and the Newton's method, we will consider applying them on the maximization of the Brockett function on $\mathrm{SO}(m)$. The two methods are first introduced, and then their convergence properties are also discussed.

2.1 Gradient and Hessian of the Brockett Function

Since Equation 1 is a constrained maximization problem on the manifold $\mathrm{SO}(m)$, these first- and second-order methods require the calculation of the (Riemannian) gradient and (Riemannian) Hessian of the Brockett function restricted on $\mathrm{SO}(m)$.

Given a matrix manifold $\mathcal{M}$ embedded in a Euclidean space $\mathbb{R}^{m\times k}$ with induced Frobenius inner product $\langle X, Y\rangle := \mathrm{Tr}\left\{X^\top Y\right\}$, the geodesic stemming from $X \in \mathcal{M}$ with tangent vector $H \in T_X\mathcal{M}$ is denoted as

$$\zeta_X(tH) \quad (t \in U), \tag{2}$$

where $U \subseteq \mathbb{R}$ is a neighborhood around 0, satisfying $\zeta_X(0) = X$ and $\dot{\zeta}_X(0) = H$. Given a *smooth* function $f : \mathcal{M} \to \mathbb{R}$, we define its differential map in the direction $H \in T_X\mathcal{M}$ as

$$\mathrm{D}f(X) : T_X\mathcal{M} \to \mathbb{R},\ H \mapsto \mathrm{D}f(X)[H] := \frac{\mathrm{d}}{\mathrm{d}t} f(\zeta_X(tH))\bigg|_{t=0}. \tag{3}$$

The second order differential map is similarly defined as

$$\mathrm{D}^2 f(X)[H,H] := \frac{\mathrm{d}^2}{\mathrm{d}t^2} f(\zeta_X(tH))\Big|_{t=0}. \tag{4}$$

As a quadratic form, the second order differential map naturally induces a symmetric bilinear form by the polarization identity:

$$\mathrm{D}^2 f(X)[H,K] := \frac{1}{4}\Big(\mathrm{D}^2 f(X)[H+K,H+K] - \mathrm{D}^2 f(X)[H-K,H-K]\Big) \quad (H,K \in T_X\mathcal{M}). \tag{5}$$

The gradient at X is then defined as the unique vector $\mathrm{grad} f(X) \in T_X\mathcal{M}$ that satisfies

$$\langle \mathrm{grad} f(X), H\rangle = \mathrm{D}f(X)[H] \quad (\forall H \in T_X\mathcal{M}). \tag{6}$$

Similarly, the Hessian is defined as the unique map $\mathrm{Hess} f(X) : T_X\mathcal{M} \to T_X\mathcal{M}$ that satisfies

$$\langle \mathrm{Hess} f(X)[H], K\rangle = \mathrm{D}^2 f(X)[H,K] \quad (\forall H,K \in T_X\mathcal{M}). \tag{7}$$

Many other formulations and approaches exist in defining the gradient and Hessian on a manifold, for further understanding of optimization on manifolds, please refer to the references [6, 4].

Let $f(x) = \mathrm{Tr}\left\{X^\mathsf{T} A X^\mathsf{T} B\right\}$ be the Brockett function defined on the manifold $\mathcal{M} = \mathrm{SO}(m)$. The tangent space at $X \in \mathrm{SO}(m)$ can be parameterized by

$$T_X\mathrm{SO}(m) = \left\{ X\Omega \,\middle|\, \Omega \in \mathbb{R}^{m\times m},\ \Omega^\mathsf{T} = -\Omega \right\} = X \cdot \mathrm{Skew}(m), \tag{8}$$

where $\mathrm{Skew}(m)$ is the set of all $m \times m$ skew-symmetric matrices. With the Frobenius metric on $\mathrm{SO}(m)$, we have its geodesic defined as

$$\zeta_X(tH) = X \cdot \exp(X^\mathsf{T} H) = X \cdot \exp(\Omega), \tag{9}$$

where $H = X\Omega \in T_X\mathrm{SO}(m)$. The gradient and Hessian are hence

$$\mathrm{grad} f(X) = X \cdot [X^\mathsf{T} A X, B] \text{ and} \tag{10}$$

$$\mathrm{Hess} f(X)[X\Omega] = -\frac{1}{2} X \cdot \left(\Big[X^\mathsf{T} A X, [B,\Omega]\Big] + \Big[B, [X^\mathsf{T} A X, \Omega]\Big] \right), \tag{11}$$

where $\Omega \in \mathrm{Skew}(m)$, and $[X,Y] = XY - YX$ is the commutator on square matrices X and Y.

2.2 Critical Points to the Brockett Function

The critical points to f satisfy $\mathrm{grad} f(X^*) = 0$, i.e., when $X^{*\mathsf{T}} A X^*$ and B commutes. But since the two matrices each has no repeating eigenvalues, the commuting condition is equivalent to the two matrices being simultaneously diagonalizable, i.e., they can be diagonalized by the same orthogonal matrices. When A and B are both diagonal matrices, the critical points are all in the form of

$$X^* = \Pi D, \tag{12}$$

where Π is a permutation matrix, and $D \in \mathrm{diag}(\pm 1, \cdots, \pm 1)$ such that $\det(\Pi D) = +1$ [1]. A total of $m! \cdot 2^{m-1}$ critical points exist, and are all isolated. The different critical points permute the diagonal elements to A.

When A and B are not diagonal, for example, $A = R_1 \Lambda_1 R_1^{\mathsf{T}}$ and $B = R_2 D_2 R_2^{\mathsf{T}}$ where D_1 and D_2 are diagonal, then the critical points are of the form

$$X^* = R_1 \Pi D R_2^{\mathsf{T}}.$$

The number of critical points does not change.

Out of the $m! \cdot 2^{m-1}$ critical points, 2^{m-1} of them are global maxima, another 2^{m-1} of them are global minima, and the rest are saddle points. To check whether a critical point is a global maxima, one checks the definiteness of the Hessian at said point. Let us assume that $A = \mathrm{diag}(\lambda_1, \cdots, \lambda_m)$ and $B = \mathrm{diag}(\mu_1, \cdots, \mu_m)$ are both diagonal, and B has strictly ascending diagonal entries ($\mu_i < \mu_j$ for $i < j$). The critical points are of the form in Equation 12. By plugging in $X^* = \Pi D$ and set $[\Omega]_{ij} = \omega_{ij}$ with $\omega_{ji} = -\omega_{ij}$ into Equation 11, one has

$$\langle \mathrm{Hess} f(X^*)[\Omega], \Omega \rangle = -\sum_{i,j} (\lambda_{\pi(i)} - \lambda_{\pi(j)})(\mu_i - \mu_j)\omega_{ij}^2, \tag{13}$$

where $\Pi^{\mathsf{T}} A \Pi = \mathrm{diag}(\lambda_{\pi(1)}, \cdots, \lambda_{\pi(m)})$ and $\pi(\cdot)$ is the permutation represented by Π. Hence, when the ordering of $\lambda_{\pi(i)}$ is the same as the ordering of μ_i – in ascending order, Equation 13 is always negative and maxima are achieved. When A has ascending diagonal elements, Π will be an identity matrix, and the maxima occur at $X^* = D$. The fact that maxima are achieved when the diagonals of A and B are of the same ordering can also be seen from the *rearrangement inequality*.

Remark 1. *(Rearrangement Inequality) For real numbers $x_1 \leq \cdots \leq x_m$ and $y_1 \leq \cdots \leq y_m$, given any permutation $\pi(\cdot)$, the following inequalities hold:*

$$x_1 y_m + \cdots + x_m y_1 \leq x_1 y_{\pi(1)} + \cdots + x_m y_{\pi(m)} \leq x_1 y_1 + \cdots + x_m y_m. \tag{14}$$

The value of the Brockett function at a critical point is equal to

$$f(\Pi D) = \mathrm{Tr}\left\{ D\Pi^{\mathsf{T}} A \Pi D B \right\} = \lambda_{\pi(1)}\mu_1 + \cdots + \lambda_{\pi(m)}\mu_m. \tag{15}$$

Hence, it matches the fact that maxima occur at points where the diagonals to $\Pi^{\mathsf{T}} A \Pi$ has the same ordering as the diagonals to B.

2.3 Gradient Ascent

Gradient ascent of a function $f(\boldsymbol{x})$ defined on a vector spaces V is implemented simply by the iteration map

$$\boldsymbol{x} \mapsto \boldsymbol{x} + \alpha \cdot \nabla f(\boldsymbol{x}), \tag{16}$$

with $\alpha = \alpha(\boldsymbol{x})$ a suitable step size scheme and $\nabla f(\boldsymbol{x})$ the gradient of f at $\boldsymbol{x}$. To apply the same scheme on manifold optimization problems, we should replace the addition in Equation 16 with the geodesic of Equation 9 as a valid translation on the manifold $\mathcal{M} = \mathrm{SO}(m)$. For the maximization to the Brockett function, we have the following iteration map as gradient ascent:

$$\begin{aligned} X_{k+1} = \varphi_{\alpha,(A,B)}(X_k) := \zeta_{X_k}\left(\alpha_k \cdot \mathrm{grad} f(X_k)\right) &= X_k \exp\left(\alpha \cdot X^{\mathsf{T}} \mathrm{grad} f(X)\right) \\ &= X_k \exp\left(\alpha_k [X^{\mathsf{T}} A X, B]\right). \end{aligned} \tag{17}$$

When the context is clear, we will simplify the notation $\varphi_{\alpha,(A,B)}$ to φ_α. To ensure convergence to the global maxima, a suitable step size scheme $\alpha = (\alpha_k)_{k\in\mathbb{N}}$ must be chosen. A constant step size scheme

$$\alpha_k = \frac{1}{4\|A\|\|B\|} \tag{18}$$

and a variable step size scheme

$$\alpha_k = \frac{1}{2\left\|[X_k^{\mathsf{T}} A X_k, B]\right\|} \ln\left(\frac{\left\|[X_k^{\mathsf{T}} A X_k, B]\right\|^2}{\|A\| \cdot \left\| \left[[X_k^{\mathsf{T}} A X_k, B], B\right] \right\|} + 1 \right) \tag{19}$$

were proposed by Moore, Mahony and Helmke [3]. More sophisticated methods, such as replacing geodesics by retractions, require deeper knowledge in differential geometry, and will not be discussed here.

Example 1. *Let us consider the case on* $\mathrm{SO}(2)$*, with*

$$A = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}, \; B = \begin{bmatrix} \mu_1 & 0 \\ 0 & \mu_2 \end{bmatrix}, \; \text{and } X = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \quad (\theta \in [0, 2\pi)).$$

The gradient is calculated to be

$$\mathrm{grad} f(X) = -\frac{1}{2}(\lambda_1 - \lambda_2)(\mu_1 - \mu_2)\sin 2\theta \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Hence, the gradient descent φ_α *becomes the following one-dimensional iteration map* $\tilde{\varphi}_\alpha$*:*

$$\begin{aligned} \varphi_\alpha(X) &= \begin{bmatrix} \cos\tilde{\varphi}_\alpha(\theta) & -\sin\tilde{\varphi}_\alpha(\theta) \\ \sin\tilde{\varphi}_\alpha(\theta) & \cos\tilde{\varphi}_\alpha(\theta) \end{bmatrix}, \\ \tilde{\varphi}_\alpha(\theta) &= \theta - \alpha \cdot \frac{1}{2}(\lambda_1 - \lambda_2)(\mu_1 - \mu_2)\sin 2\theta. \end{aligned} \tag{20}$$

Suppose $\alpha > 0$ *for maximization and* $\lambda_1 < \lambda_2$*,* $\mu_1 < \mu_2$*. By analyzing* $|\tilde{\varphi}'_\alpha| < 1$*, one can see that attractive stationary maxima exist at* $\theta = 0$ *and* π *for* $\alpha(\lambda_1 - \lambda_2)(\mu_1 - \mu_2) \in (0, 2)$*.*

2.3.1 Convergence Analysis

definition, convergence, constant step scheme and variable step scheme

2.4 Newton's Method

2.4.1 Convergence Analysis

Quadratic convergence

2.4.2 Solution to Newton's Equation

For the Brockett function, we define the Newton's method by the following Newton's equation:

$$\begin{gathered} \mathrm{Hess} f(X)[X\Omega] = -\mathrm{grad} f(X) \\ \frac{1}{2}\left(\left[X^\top AX, [B, \Omega]\right] + \left[B, [X^\top AX, \Omega]\right]\right) = [X^\top AX, B], \end{gathered} \tag{21}$$

where we solve for the Newton's step $\Omega \in \mathrm{Skew}(m)$. The evolution will be of the form $X \mapsto X\exp(\Omega)$. Unlike what is believed by other authors [3], Equation 21 might not have a solution. To see this, we can construct a counter example in SO(2): consider A and B to be 2×2 diagonal matrices, there exists no solution to the Newton's equation above at

$$X = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}.$$

The counter example can be further extended to higher dimensions by setting A and B to be diagonal, and take

$$X = \left[\begin{array}{cc:c} 1/\sqrt{2} & -1/\sqrt{2} & \\ 1/\sqrt{2} & 1/\sqrt{2} & \\ \hdashline & & \mathbb{1}_{m-2} \end{array}\right],$$

with $\mathbb{1}_n$ being an $n\times n$ identity matrix. Therefore, the existence of a solution to Equation 21 is not trivial.

If a unique solution does exist, however, we denote it by $\Omega = W(A,B,X)$. The function $W(A,B,X)$ can be implemented numerically in a computer by setting

$$\Omega = \sum_{i=1}^{m-1}\sum_{j=i+1}^{m} \omega_{ij} L_{ij},\ [L_{ij}]_{k,l} = \begin{cases} +1 & , k=i, l=j \\ -1 & , k=j, l=i \\ 0 & , \text{otherwise.} \end{cases} \tag{22}$$

with $\omega_{ij}\in\mathbb{R}$ and L_{ij} a basis to the anti-symmetric matrices. Then, Equation 21 can be represented as a matrix multiplied on the vectorized coefficients ω_{ij}'s. It can be solved by simply multiplying the inverse matrix. The Newton's method is the following iterative map:

$$X_{k+1} = \psi_{(A,B)}(X_k) := \zeta_{X_k}\left(X_k \cdot W(A,B,X_k)\right) = X_k \cdot \exp\left(W(A,B,X_k)\right). \tag{23}$$

When the context is clear, we will write the iteration map as ψ instead of $\psi_{(A,B)}$ for its simplicity.

Example 2. *Continue with the setup of Example 1 and let*

$$\Omega = \begin{bmatrix} 0 & \omega \\ -\omega & 0 \end{bmatrix}.$$

Solving the Newton's equation, we obtain that

$$W(A,B,X) = \omega \cdot \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \ \omega = \frac{\tan 2\theta}{2}.$$

Hence, the Newton's method ψ becomes the following one-dimensional iteration map $\tilde{\psi}$:

$$\psi(X) = \begin{bmatrix} \cos\tilde{\psi}(\theta) & -\sin\tilde{\psi}(\theta) \\ \sin\tilde{\psi}(\theta) & \cos\tilde{\psi}(\theta) \end{bmatrix},$$
$$\tilde{\psi}(\theta) = \theta - \omega = \theta - \frac{\tan 2\theta}{2} \mod 2\pi. \tag{24}$$

The iteration map $\tilde{\psi}$ converges quadratically to $\theta = 0$, $\pi/2$, π, and $3\pi/2$. One also sees that the Newton's step does not exist for $\theta = \pi/4$, $3\pi/4$, $5\pi/4$ and $7\pi/4$.

Remark 2. *The solution to the Newton's equation can also be implemented using matrix pseudo-inverse: given the vectorization operator* $\mathrm{vec}(\cdot)$ *defined as stacking the columns of a matrix into a single vector. We have, for $m \times m$ matrices A, X, B of same size,*

$$\mathrm{vec}(AXB) = (B^\top \otimes A)\mathrm{vec}(X), \tag{25}$$

where $\otimes$ is the Kronecker product. Then the commutator matrix K is defined as the permutation

$$\mathrm{vec}(X^\top) = K\mathrm{vec}(X). \tag{26}$$

The Newton's equation can be transformed into a set of vectorized equations: let us define $\mathrm{ad}_A := \mathbb{1}_m \otimes A - A^\top \otimes \mathbb{1}_m$ *as the vectorization of the commutator (adjoint) operator $[A,\cdot]$, and $\{A,B\} := AB + BA$ as the matrix anti-commutator, then we have*

$$\frac{1}{2}\left\{\mathrm{ad}_{X^\top AX}, \mathrm{ad}_B\right\}\mathrm{vec}(\Omega) = \mathrm{vec}\left([X^\top AX, B]\right), \tag{27}$$
$$K\mathrm{vec}(\Omega) = -\Omega, \tag{28}$$

where the second equation enforces the skew-symmetric property of $\Omega \in \mathrm{Skew}(m)$*. By combining the two equations above, we can solve for Ω via:*

$$\mathcal{M}\mathrm{vec}(\Omega) := \begin{bmatrix} \frac{1}{2}\left\{\mathrm{ad}_{X^\top AX}, \mathrm{ad}_B\right\} \\ K + \mathbb{1}_{m^2} \end{bmatrix} \mathrm{vec}(\Omega) = \begin{bmatrix} \mathrm{vec}\left([X^\top AX, B]\right) \\ 0 \end{bmatrix} =: Z \tag{29}$$

$$\to \mathrm{vec}(\Omega) = \mathcal{M}^{+} Z, \tag{30}$$

where $\mathcal{M}^{+}$ is the matrix pseudo-inverse to $\mathcal{M}$.

The result obtained from applying the pseudo-inverse coincides with that obtained by solving the linear equations on ω_{ij}'s when the solution to the latter exists. However, the solution set of the former is larger than that of the latter.

newton-type method, Newton's method (need to check to comply with normal terms and usage), quadratic convergence, solution to newton's equation, fractal like structure,

3 Basins of Attraction to Newton's Method

3.1 Symmetries to Newton's Step

Theorem 3.1. *For the unique $m \times m$ skew-symmetric matrix $\Omega = W(A, B, X)$ that satisfies the Newton's equation of Equation 21, we have, for any R_1 and $R_2 \in \mathrm{SO}(m)$,*

1. $W(R_1^\intercal A R_1, R_2^\intercal B R_2, X) = R_2^\intercal W(A, B, R_1 X R_2^\intercal) R_2$,
2. $W(B, A, X) = -X^\intercal W(A, B, X^\intercal) X$.

Proof. Both can be shown by simply equating the two sides to the equations. ∎

The properties of W listed above translates directly to the symmetry in the Newton's map ψ.

Theorem 3.2. *Consider diagonal matrices A, B and any X, R_1, $R_2 \in \mathrm{SO}(m)$, the Newton's map $\psi_{(A,B)}$ and $\psi_{(R_1^\intercal A R_1, R_2^\intercal B R_2)}$ has the following relation:*

$$\psi_{(R_1^\intercal A R_1, R_2^\intercal B R_2)}(X) = R_1^\intercal \cdot \psi_{(A,B)}(R_1 X R_2^\intercal) \cdot R_2. \tag{31}$$

Proof. The proof is trivial by considering the identity

$$\exp(R^\intercal X R) = R^\intercal \exp(X) R$$

for all $R \in \mathrm{SO}(m)$. ∎

From the third property of Theorem 3.2, we know that without loss of generality, the behavior of ψ to any (A, B) is related by rotations to the behavior of (A, B) both being diagonal. Henceforth, if it is not stated explicitly, we will be working with $A = \mathrm{diag}(\lambda_1, \cdots, \lambda_m)$ and $B = \mathrm{diag}(\mu_1, \cdots, \mu_m)$ being diagonal with λ_i's and μ_i's having the same ordering.

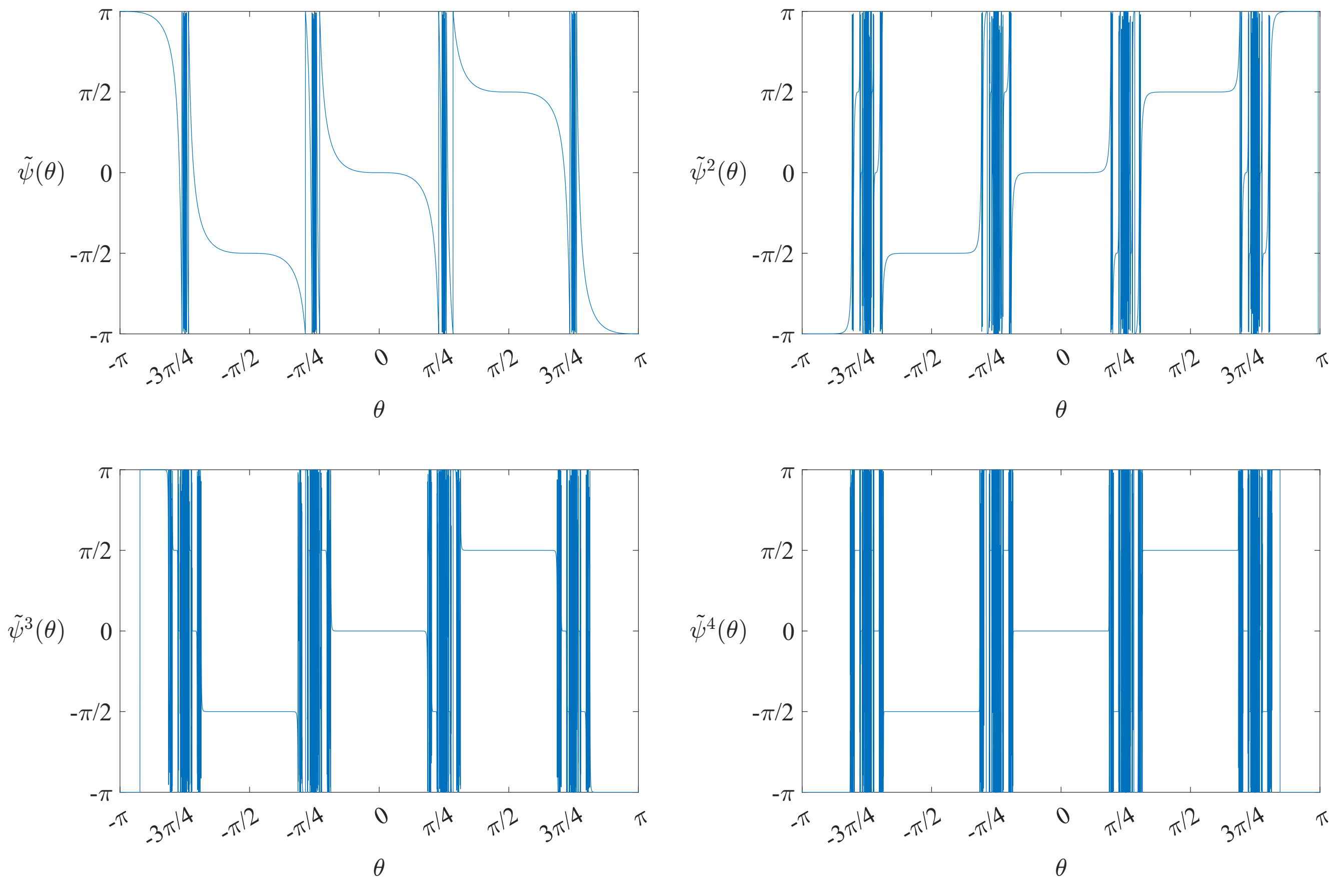

Figure 1: Newton's Iteration on SO(2), represented by the parameter $\theta \in (-\pi, \pi]$.

3.2 Basins of Attraction on SO(2)

Let us consider again the case on SO(2) as in Example 2 since it can be analyzed analytically. The matrix X_k parameterized by θ_k is updated by the iterative map $\theta_{k+1} = \tilde{\psi}(\theta_{k+1})$. A few iteration results are shown in Figure 1. For clarity of the figures, the range to the modulo 2π in Equation 24 is $(-\pi, \pi]$. A one-dimensional iterative maps can generate self-similar fractal structures [cite], and when zooming in to the details around $\pm\pi/4$ and $\pm 3\pi/4$, similar staircase structures can be seen.

Let us define the basin of attraction to the critical point $\theta^* = 0, \pm\pi/2$, or π as

$$\mathcal{B}_{\theta^*}(\tilde{\psi}) = \left\{ \theta \in (-\pi, \pi] \,\middle|\, \tilde{\psi}^n(\theta) \to \theta^* \text{ as } n \to \infty \right\}. \tag{32}$$

The connected component of $\mathcal{B}_{\theta^*}(\tilde{\psi})$ to θ^* is denoted as $\mathcal{B}^{\circ}_{\theta^*}(\tilde{\psi})$. For the one-dimensional iteration map $\tilde{\psi}$, since we have the symmetry of

$$\tilde{\psi}(-\theta) = -\tilde{\psi}(\theta),$$

the connected component of the basin of attraction to $\theta^* = 0$ is also symmetric to $\theta = 0$, and it can be easily characterized by

$$\theta \in \mathcal{B}_0^\circ(\tilde{\psi}) = (-\rho_0, \rho_0) \Rightarrow \left|\tilde{\psi}(\theta)\right| < |\theta| \,,$$

with ρ_0, the boundary to $\mathcal{B}_0^\circ(\tilde{\psi})$, satisfying the equality

$$\rho_0 = -\left(\rho_0 - \frac{\tan 2\rho_0}{2}\right) \Rightarrow \rho_0 = 0.582780592\cdots\,. \tag{33}$$

Next, by the symmetry to the Newton's step where

$$\tilde{\psi}(\theta + \pi/2) = \tilde{\psi}(\theta) + \pi/2,$$

the connected component of the basins of attraction to $\theta = \pm\pi/2$ and π is obtained simply by translating $\mathcal{B}_0^\circ(\tilde{\psi})$.

3.3 General Properties of the Basins

The shape to the basins of attraction is highly correlated with the behavior of $W(A, B, X)$. Let us apply the results in Section 3.1 to give certain descriptions on the symmetry to the basins of attraction on $\mathrm{SO}(m)$. We only need to consider the case where both A and B are diagonal, and all the other cases follow suit in accordance with Theorem 3.2.

Consider any $D \in \mathrm{diag}(\pm 1, \cdots, \pm 1)$, we have

$$D^\intercal A D = A \text{ and } D^\intercal B D = B.$$

Thus, by restricting $X \mapsto XD \in \mathrm{SO}(m)$ and setting $R_1 = \mathbb{1}_m$ and $R_2 = D$ in Theorem 3.2, we have

$$\psi(XD) = \psi(X)D. \tag{34}$$

Therefore, if $X \to X^*$, then $XD \to X^*D$ for X^* and X^*D both a critical point, and D as defined in

3.4 Basins of Attraction on SO(3)

We can visualize the basins of attraction to the $24 = 3! \cdot 2^{3-1}$ critical points on $\mathrm{SO}(3)$ by displaying the result on its Lie algebra, i.e., $\mathfrak{so}(3)$, which is isomorphic to $\mathbb{R}^3$. Denote the $\Theta = [x, y, z] \in$

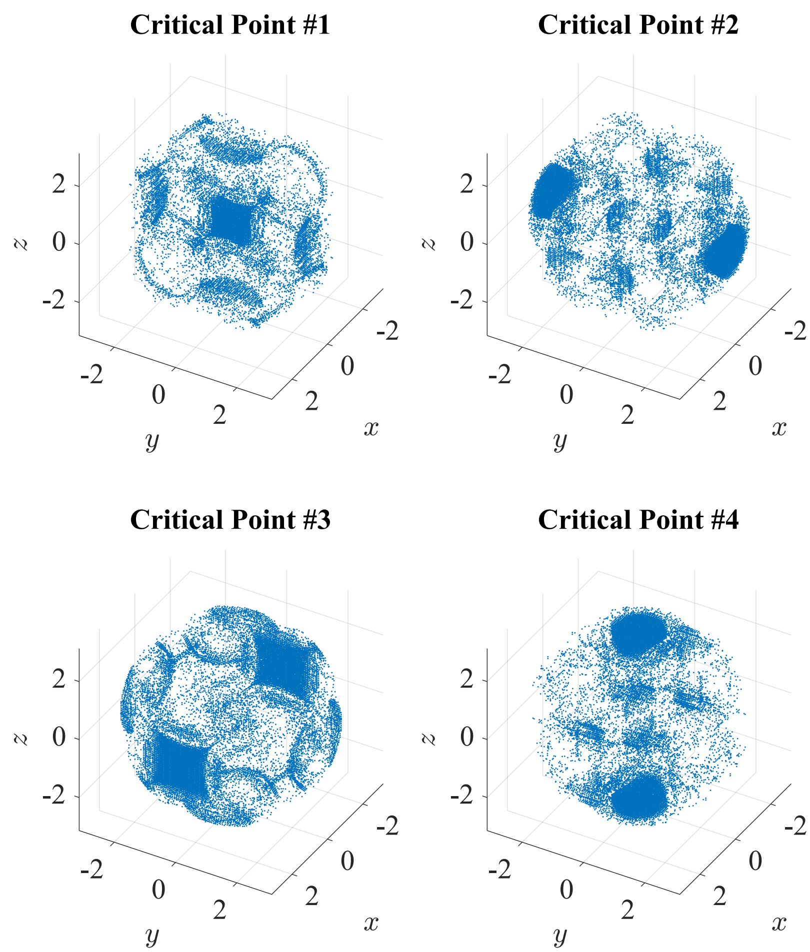

Figure 2: Basins of attraction to the maxima of the Brockett function on SO(3).

$\mathfrak{so}(3) \cong \mathbb{R}^3$ with $\|\Theta\| \leq \pi$ such that $X = \exp(\Theta)$, then we have the following relations:

$$\Theta = \frac{\theta}{2\sin\theta}(X - X^{\mathsf{T}}),\ \theta = \arccos\frac{\operatorname{Tr}\{X\} - 1}{2}, \tag{35}$$

$$X = \exp(\Theta) = \exp(xL_x + yL_y + zL_z), \tag{36}$$

where

$$L_x = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, L_y = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, L_z = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{37}$$

are the bases to $\mathfrak{so}(3)$. Geometrically, since any rotation in three-dimensional space can be identified by an axis $\boldsymbol{n}$ (with $\|n\| = 1$) and an angle $\theta \in [0, \pi]$, they can be combined to form $\Theta = \theta\boldsymbol{n}$.

The symmetry to the basins of attraction for SO(3) is much more complicated. A simple visualization to its four basins of attraction to the maxima (critical points 1 to 4), viz., $X_1^* = \mathbb{1}_3$,

$X_2^* = \mathrm{diag}(1, -1, -1)$, $X_3^* = \mathrm{diag}(-1, 1, -1)$, and $X_4^* = \mathrm{diag}(-1, -1, 1)$, are shown in Figure 2. Though the shapes look wildly different, they actually have the same volume with respect to the Haar measure on SO(3) (or respectively, on $\mathfrak{so}(3)$).

4 Gradient Ascent Followed by Newton's Method

The two line search methods introduced in Section 2 have their advantages and disadvantages: gradient ascent has great global convergence property, but ill-selected step sizes easily result in stagnation; Newton's method has local quadratic convergence at extrema, but globally, it can either converge to saddle points or diverge! The two methods have opposite performances globally and locally. It is hence of normal practice that one combine the two methods: first, use gradient ascent to reach near one of the global maxima, then, apply Newton's method to quickly converge to the maximum. This is the first method on speeding up the line search methods on the maximization of the Brockett function.

A natural question would be, given gradient ascent φ_α and Newton's method ψ as described in Equation 17 and Equation 23, what is the optimal combination of step size α and number of gradient ascents n so that

$$\mathrm{Pr}_{X\sim\mathrm{Unif}(\mathrm{SO}(m))}\left\{\varphi_\alpha^n(X) \in \mathcal{B}_{\min}(\psi)\right\} \geq 1 - \varepsilon, \tag{38}$$

where $\varepsilon \in [0, 1]$ is the acceptable error. The uniform distribution is with respect to the Haar measure on SO(m). The smaller the step size is, the more iterations are needed; the less iterations are wanted, the larger the step size should be – a trade-off between the two variables is apparent. Hence, by optimal, we meant that given a number α^*, under the condition that $\alpha \leq \alpha^*$, the smallest n is chosen such that Equation 38 holds. The role of the upper bound α^* on the step size will be explained later on.

In this section, we first consider the Brockett function on SO(2) as an example to shed light on the relation between α^* and optimal (α, n)-pairs. Then, we will give bounds on the case of SO(3) and further generalize the results onto SO(m).

4.1 On SO(2)

We first analyze the case on SO(2) since α^* and the optimal (α, n)-pair can be calculated.

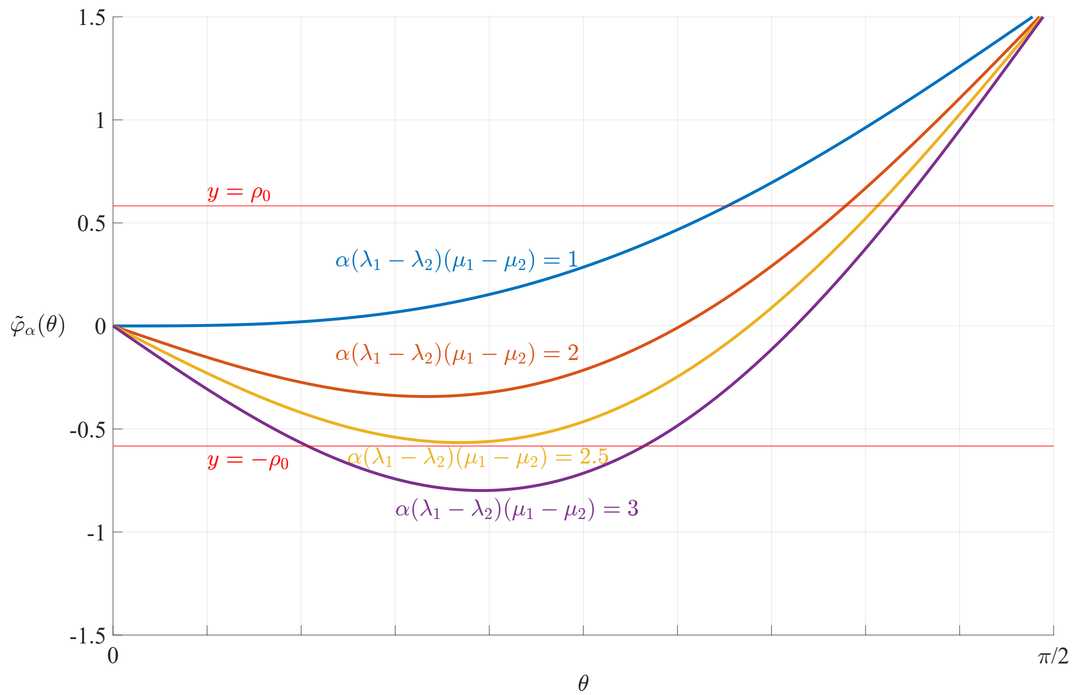

Figure 3: A single gradient step on SO(2).

A simpler formulation would be to ask for a computable lower bound on the left hand side to Equation 38. A natural choice would be to replace $\mathcal{B}_{\min}(\psi)$ by $\mathcal{B}^\circ_{\min}(\psi)$, then we can utilize Equation 33. We can now calculate

$$\mathrm{Pr}_{X\sim\mathrm{Unif}(\mathrm{SO}(2))}\left\{\varphi^n_\alpha(X) \in \mathcal{B}^\circ_{\min}(\psi)\right\} = \mathrm{Pr}_{\Theta\sim\mathrm{Unif}([0,\pi])}\left\{\tilde{\varphi}^n_\alpha(\Theta) \in [0,\rho_0)\right\}, \tag{39}$$

where $\tilde{\varphi}_\alpha$ is as defined in Example 1. As can be seen in Figure 3, in order to maximize Equation 39 while not resulting in fractal structures (for large α's), we need the first extrema of $\tilde{\varphi}_\alpha$ to have absolute value less than ρ_0. Hence, we need

$$\begin{aligned} \theta - \alpha^*(\lambda_1-\lambda_2)(\mu_1-\mu_2)\frac{\sin 2\theta}{2}\bigg|_{\cos 2\theta = \frac{1-\rho_0}{\alpha^*(\lambda_1-\lambda_2)(\mu_1-\mu_2)}} &= -\rho_0 \\ \to \alpha^*(\lambda_1-\lambda_2)(\mu_1-\mu_2) &= 2.536558\cdots. \end{aligned} \tag{40}$$

Given (α, n) with $0 < \alpha < \alpha^*$, one can immediately calculate Equation 39 by finding the first intercept of $y = \tilde{\varphi}^n_\alpha$ with $y = \rho_0$. A few values to the left hand side of Equation 38 are shown in Figure 4. One can see that when $\alpha > \alpha^*$, the probability of converging to the global minima quickly degrades or turns chaotic. The behavior to $\tilde{\varphi}_\alpha$ when α is large is not considered here, since the complicated fractal structure inherent makes the analysis on the probability very difficult.

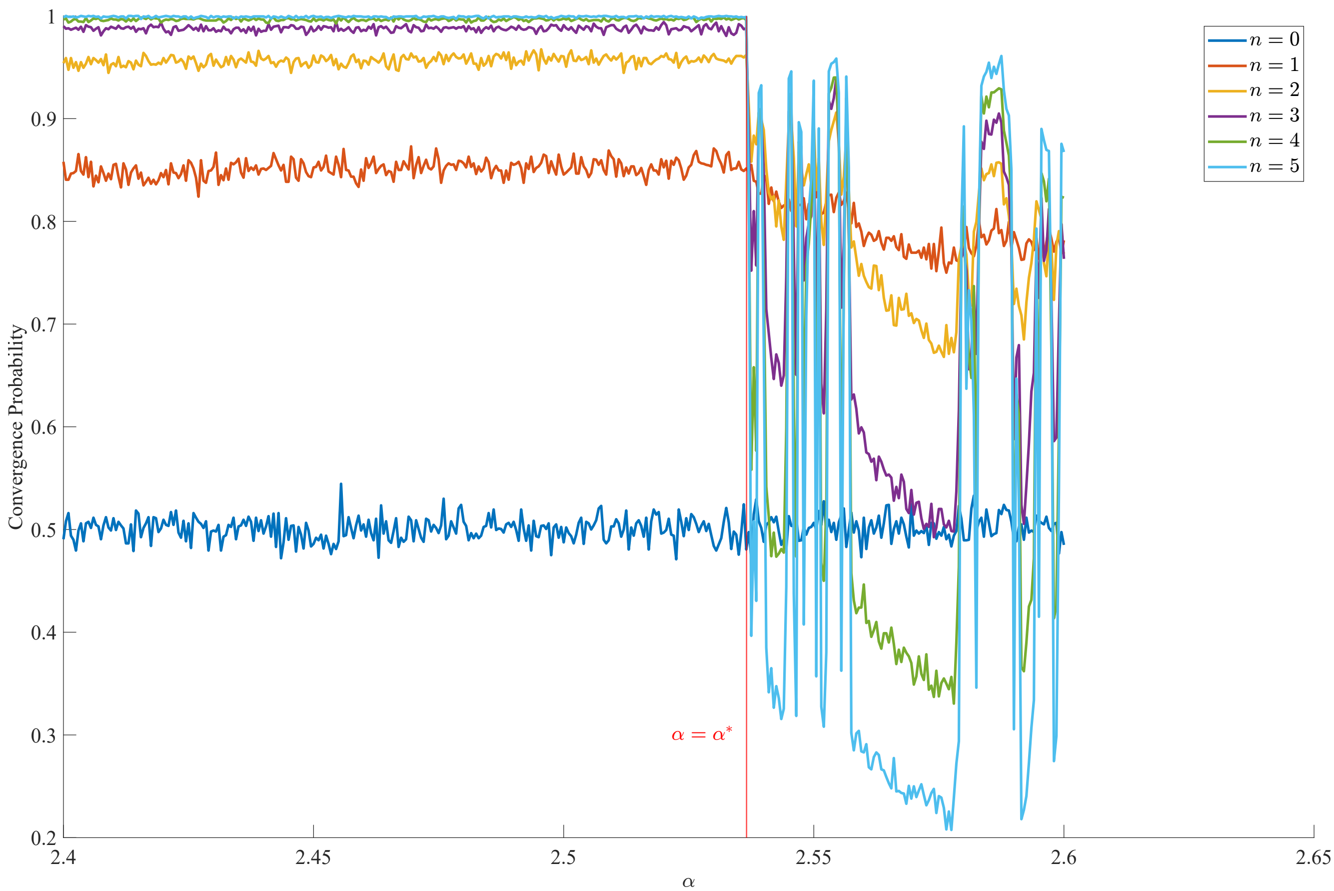

Figure 4: Convergence probability of (α, n)-pair on SO(2), with parameters $A = B = \text{diag}(1, 2)$.. Each sample point ran over 2000 samples of optimization with random initial condition.

Amazingly, since Equation 40 only depends on $(\lambda_1 - \lambda_2)$ and $(\mu_1 - \mu_2)$, the maximum step size α^* can easily be calculated for designing a suitable $\alpha < \alpha^*$, and only $n = 4$ is required to ensure 99% convergence irrespective of the A and B given.

4.2 On SO(3)

For the case of SO(3), the problem is three-dimensional and becomes way harder to solve. Besides having α and n as a parameter, the diagonal elements chosen for A and B also has a major nontrivial effect on the convergence probability. This is different from the case on SO(2), where α^* is simply inversely proportional to $(\lambda_1 - \lambda_2)(\mu_1 - \mu_2)$. Simulated results on the effect of the said parameters are shown in Figure 5, Figure 6, and Figure 7. The data for α^*'s in the last figure are listed in Table 1.

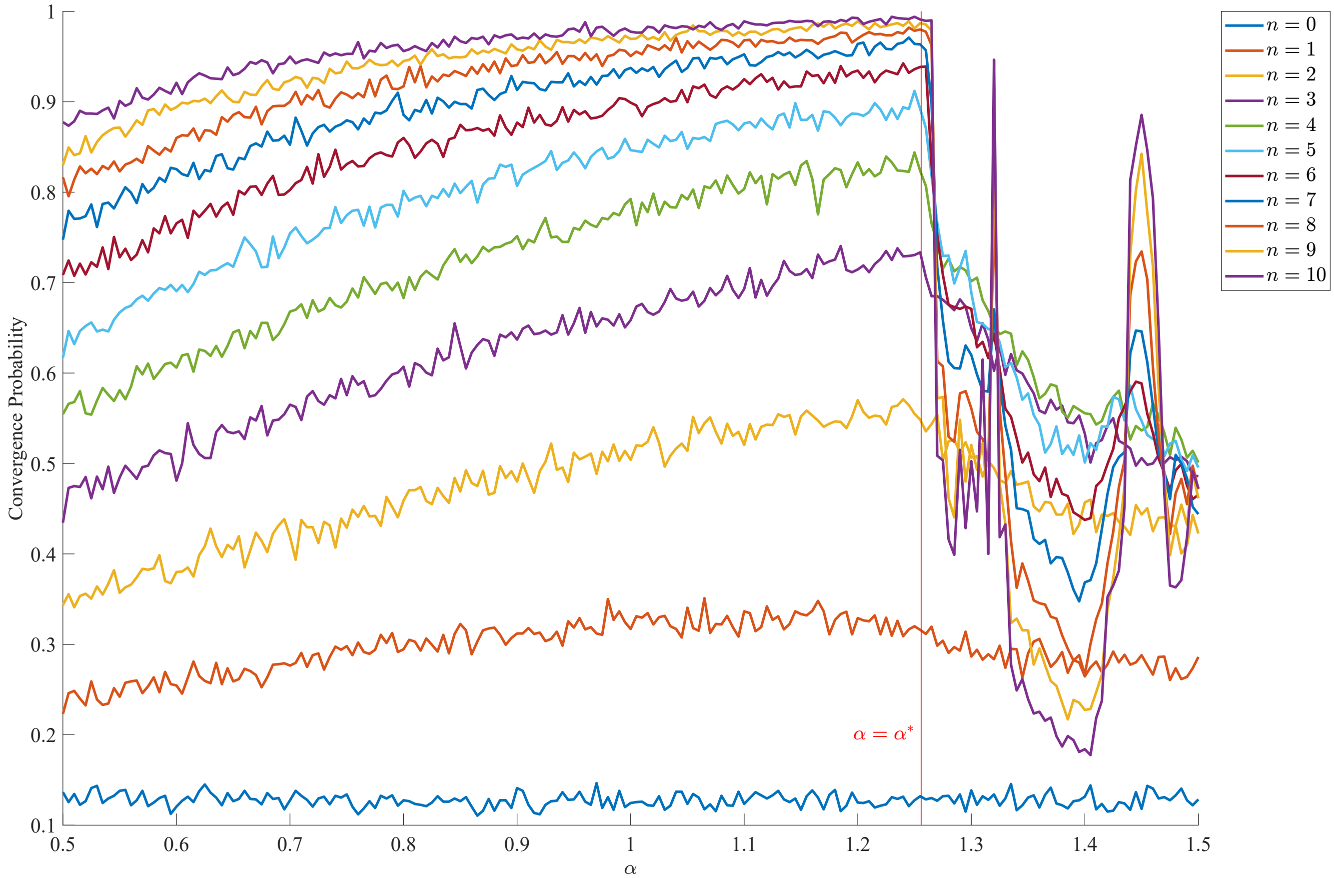

Figure 5: Convergence probability of (α, n)-pair on SO(3), with parameters $A = B = \mathrm{diag}(1, 2, 3)$. Each sample point ran over 2000 samples of optimization with random initial condition. The exact value to α^* is unknown except that it exists.

An experience formula for α^* is: denote $\lambda_{ij} = \left|\lambda_i - \lambda_j\right|$ and $\mu_{ij} = \left|\mu_i - \mu_j\right|$, then

$$\alpha^* = \frac{2.536558\cdots}{(\lambda_{12}\lambda_{23}\lambda_{31}\mu_{12}\mu_{23}\mu_{31})^{1/2}}. \tag{41}$$

4.3 Generalizing to SO(m)

5 Method of Population

In Section 3, we discuss the fractal-like structure to the basins of attraction of extrema under Newton's method ψ, where we denote $\mathcal{B}_X(\psi)$ as the basin of attraction to the extrema X.

Is there an arrangement to a set of initial points, such that no matter the A and B given, at least one of the initial points lies in the basins of attraction to the global maxima under the

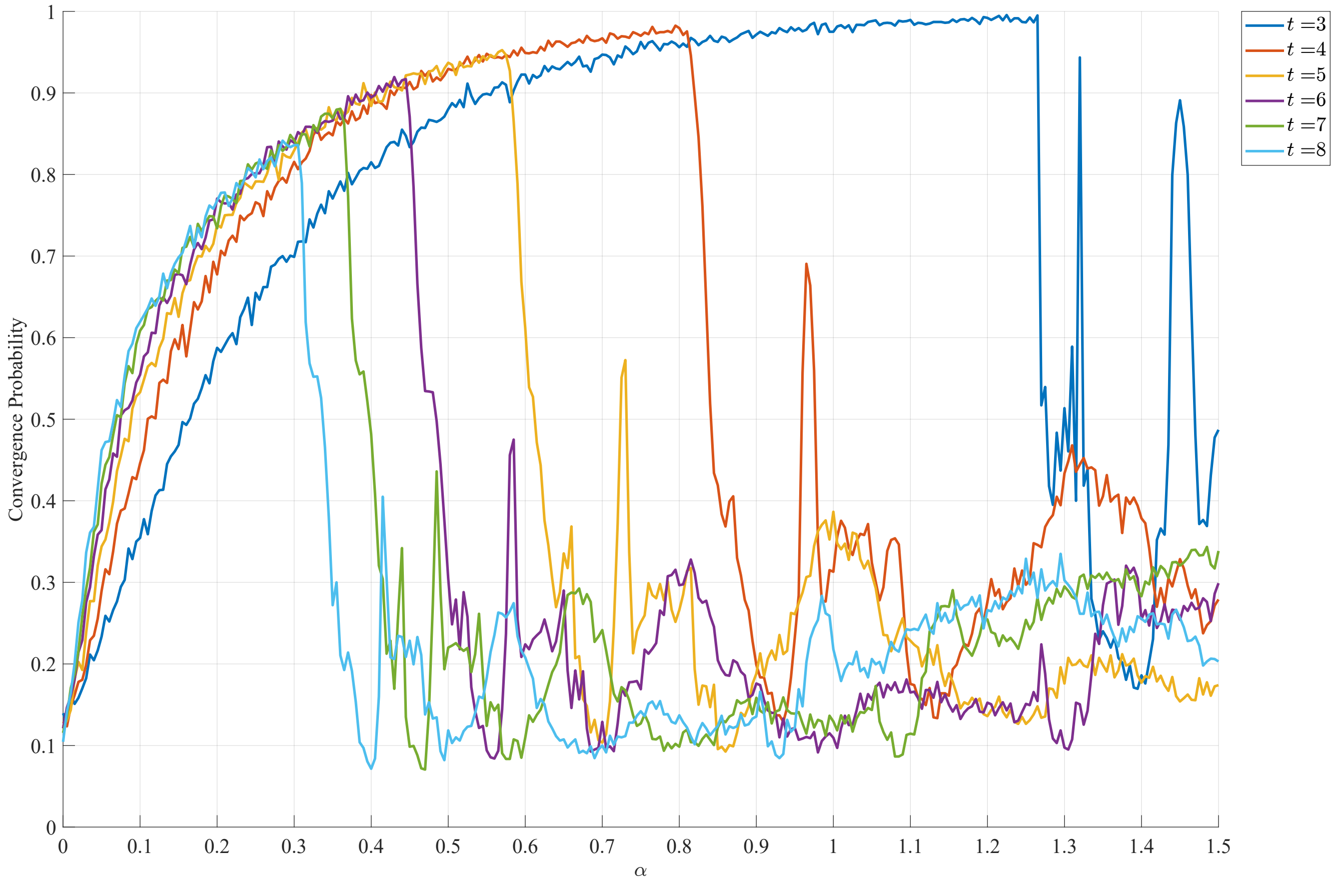

Figure 6: Convergence probability of $(\alpha, 10)$-pair on SO(3), with parameters $A = \text{diag}(1, 2, t)$ and $B = \text{diag}(1, 2, 3)$. Each sample point ran over 2000 samples of optimization with random initial condition.

Newton's method as described by Equation 23? This is the idea of *population methods*, where a given set of points called a *population* ensures the convergence of at least one of the initial points, called *individuals*, to the global maxima.

The setup to population methods requires a given arrangement to the population. Normally, the population is given by random sampling over a uniform, Gaussian or Cauchy distribution. However, with the knowledge to the shape of the basins of attraction over SO(m), deterministic arrangements can be given to ensure convergence with probability one.

Example 3. *Let us start our investigation by an example in the 2-dimensional case.*

Homotopy methods on finding the zeros to a function can ensure probability-one convergence to maxima. The shape of the basins of attraction will change as A and B changes

α^* t_2 / t_1	3	4	5	6	7	8
3	1.26617	0.809009	0.591441	0.458058	0.373874	0.314314
4	0.812012	0.510811	0.376076	0.295646	0.242793	0.204605
5	0.576476	0.371171	0.273273	0.216366	0.178378	0.151351
6	0.445045	0.289189	0.214715	0.17012	0.140541	0.11972
7	0.364364	0.234234	0.176977	0.14039	0.116216	0.0988989
8	0.307107	0.196396	0.14965	0.119469	0.0990991	0.0840841

Table 1: Relation of α^* with t_1 and t_2 from Figure 7.

6 Conclusion

We proposed three novel methods in analyzing

Hard analysis on basins of attraction to Newton's method can be replaced by analyzing quasi-Newton methods or approximate Newton's method.

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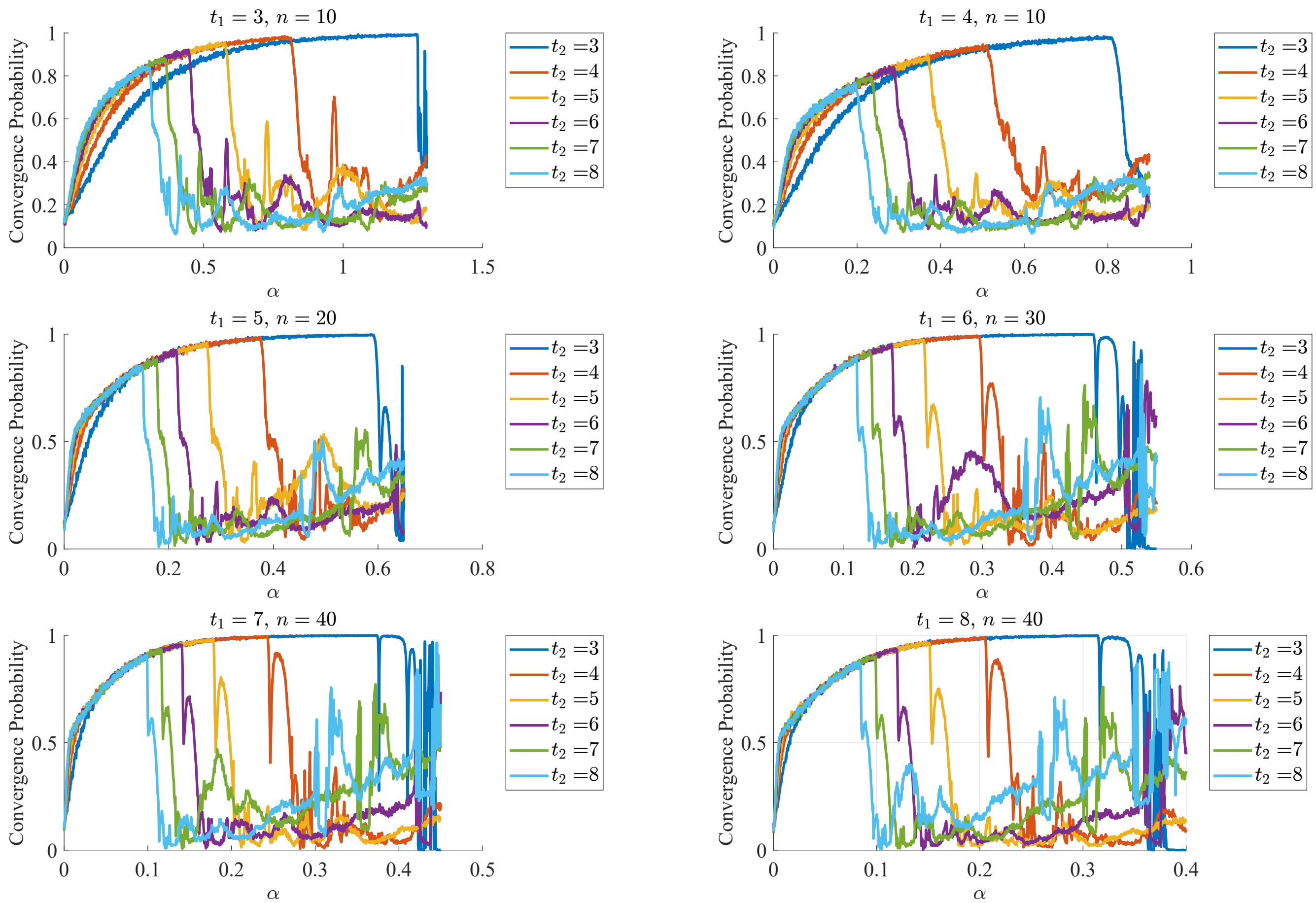
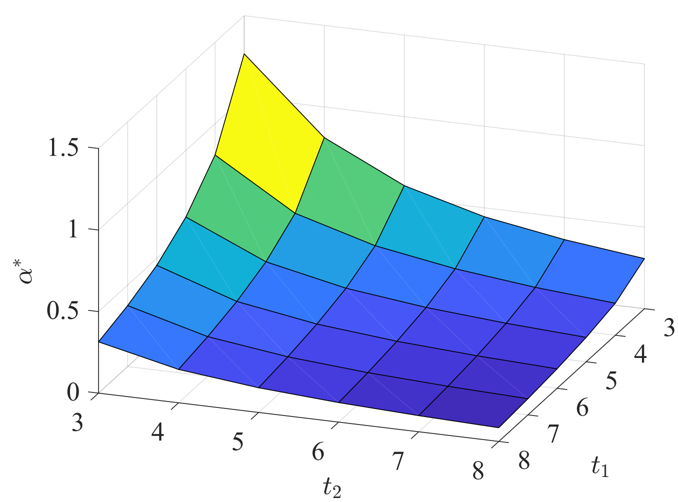


(a) Convergence probability with different parameters A and B. Each range of α for a given t_1 is divided into 1000 sample points. And each sample point ran over 2000 samples of optimization with random initial condition.



(b) Relation of α^* with t_1 and t_2.

Figure 7: Convergence probability and the respective optimal step size α^* with parameters $A = \text{diag}(1, 2, t_1)$ and $B = \text{diag}(1, 2, t_2)$.